

Why SQL Should Be Used for Data Analysis



A basic knowledge of SQL and its query language is essential for any aspiring data analyst and data scientist. This article discusses SQL and its data analysis features.

Structured Query Language (SQL) is a must for a data analyst or data scientist. SQL provides the basic storage and retrieval platform for organisations working with data. It provides rich data retrieval and report generation facilities, along with numerous data storage and security features. Though it is a complex multi-layered system, it is still simple and easy to understand. This is why data analysts are using SQL to understand and explore data before moving to higher levels of analysis and visualisation.

What is SQL and why is it so popular?

SQL is a relational database system and is used to create, store and process data. But for a data analyst, it is better to compare SQL with a spreadsheet, particularly with Microsoft's Excel. Both SQL and Excel are simple and robust for data storage, but there are differences when it comes to handling them. Excel is basically a graphical user interface and performs data

visualisation in a very simple way. It provides a shallow data analysis platform (only one workbook) with the graphical visualisation facility. Moreover, it is not meant for high volumes of data. SQL, on the other hand, is very rich in data analysis, especially in report generation. It supports very complex analysis (queries) combining multiple tables and databases. SQL also supports analysis

in different forms, like interactive and stored (in the form of stored procedures and virtual tables (VIEW)) queries.

Excel is suitable for smaller data sets, and becomes slow and inefficient when the data is filtered with complex formulas. This performance degradation is quite high for a data sheet with only a few thousand rows. In SQL, even 10 million rows can be processed fairly quickly.

Table 1: Comparing the data analysis capabilities of Excel and SQL

Excel	SQL
Stores data in sheets	Stores data in tables
Maintains no relationship among data distributed in sheets	Maintains relationships among tables
Works on a single table (sheet)	Can easily be implemented on multiple tables
Supports limited query options	Supports extensive query options
Queries cannot be stored	Queries can be stored as VIEW and STORED PROCEDURE
Visualisation of data and its analysis is relatively simple	Visualisation of data in graphical form is absent